

Water on 17 asteroids

Japanese astronomers have discovered evidence of water on 17 asteroids using data from AKARI, an infrared satellite. <https://digit.in/jan19-21>



Human hearing for AI

Russian scientists are closer to creating a digital system with the ability to process speech in real-life sound environment. <https://digit.in/jan19-22>

BLOCKCHAIN-BASED PRIVACY SYSTEMS

A distributed ledger or public blockchain, such as Ethereum, has completely different traits from the internet. It could safely be seen as the exact opposite of the data storage methods used by the internet. Instead of making copies of data and transferring information to other computers with a simple copy-paste method, blockchain facilitates a transfer of ownership of the information or data. Blockchain provides immutability and transparency in data storage and usage which gives the users' right to privacy back to them. The technology, however, is still in its infancy but promises better online privacy.

HOW IT WORKS

Blockchain technology features no central location to store data, unlike the internet, therefore negating a single point of attack for hackers and other agencies. Fortunately, this does not just apply to cryptocurrency transactions as is sometimes the notion. There are multiple companies that are building off this secure technology and decentralising data. Encrypted messaging, tracking enterprise documents and creation of legal documents with ease are some of its applications.

Blockchain makes it possible to store assets and data in a way that negates the possibility of duplication without permission. It also always allows the source to be known if needed. The immutable records are stored and distributed across all users, each of whom has their own private cryptographic key. Alongside, the active development of a 'zero-knowledge proof system', that incorporates both blockchain technology and cryptography, allows users to verify a claim without disclosing any data at all. This could possibly be the pinnacle of online privacy and is being researched and developed thoroughly.

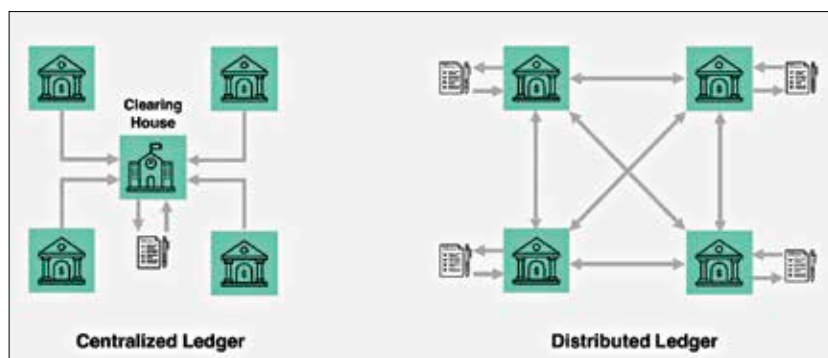
Blockchain sports a public-private architecture which defines and refines the privacy of digital data. Reiterating the previous example, blockchain tech-

nology exempts users from having to share personal information like their date of birth over and over. Instead, there is one permanent record of this information on the blockchain and then the owner can grant temporary access to the official record whenever and wherever it is needed.

In the system of blockchain-based privacy systems, public information including tax records, government records, local school budgets and more can be kept public and open-source, while personal data of citizens and sensitive data of companies can be kept private and secure. The current digital economy can be turned on its head with the adoption of the block-

noticeable in this area after they began their E-residency program which allows the citizens to record their data on the blockchain. Undeniably, mass adoption also requires awareness. Programs such as Mainframe and many others have been working towards disintegrating the popular opinion that blockchain technology mainly caters to cryptocurrency investment and solidifying the technology's real-world applications in the citizens' minds.

Additionally, numerous leading venture capital firms from all over the world like FBG Capital, Rockaway Blockchain Capital and more are recognising the merits of blockchain technology, especially in the online



Distributed ledger technology (blockchain) versus centralised ledger technology (internet).

chain. Today, citizens freely give away their data, many times unknowingly, to companies like Facebook, who in turn monetise this data without sharing any revenues. However, in a blockchain system, the data is truly our own and the ability to access this data lies solely with us and who we deem fit. Therefore, users can now their share their data knowingly and willingly and possibly even monetise it by reaping benefits or premium access to services in exchange for their highly-coveted data.

ADOPTION OF BLOCKCHAIN TECHNOLOGY

The adoption of blockchain technology has been tediously slow. However, countries like Singapore, Philippines, Switzerland have slowly adopted policies that support blockchain technology and privacy. Estonia has also been

privacy sector. They are investing in and supporting many projects that involve privacy to produce marketable goods and products that hand the power back to the consumer. JPMorgan Chase launched a new blockchain payment processing network with the Royal Bank of Canada and Australia and New Zealand Banking Group.

Indian banks and other financial entities are also heavily investing in this new technology which shields customer data from breaches or leaks and offers a much higher level of security. The technology is quite new in India and still has ways to go before it can be implemented in substantial ways to ensure better data protection. However, the foundation is being built slowly. Blockchain-based privacy systems could lead the charge in eliminating the distressing concerns of people when it comes to data privacy. 